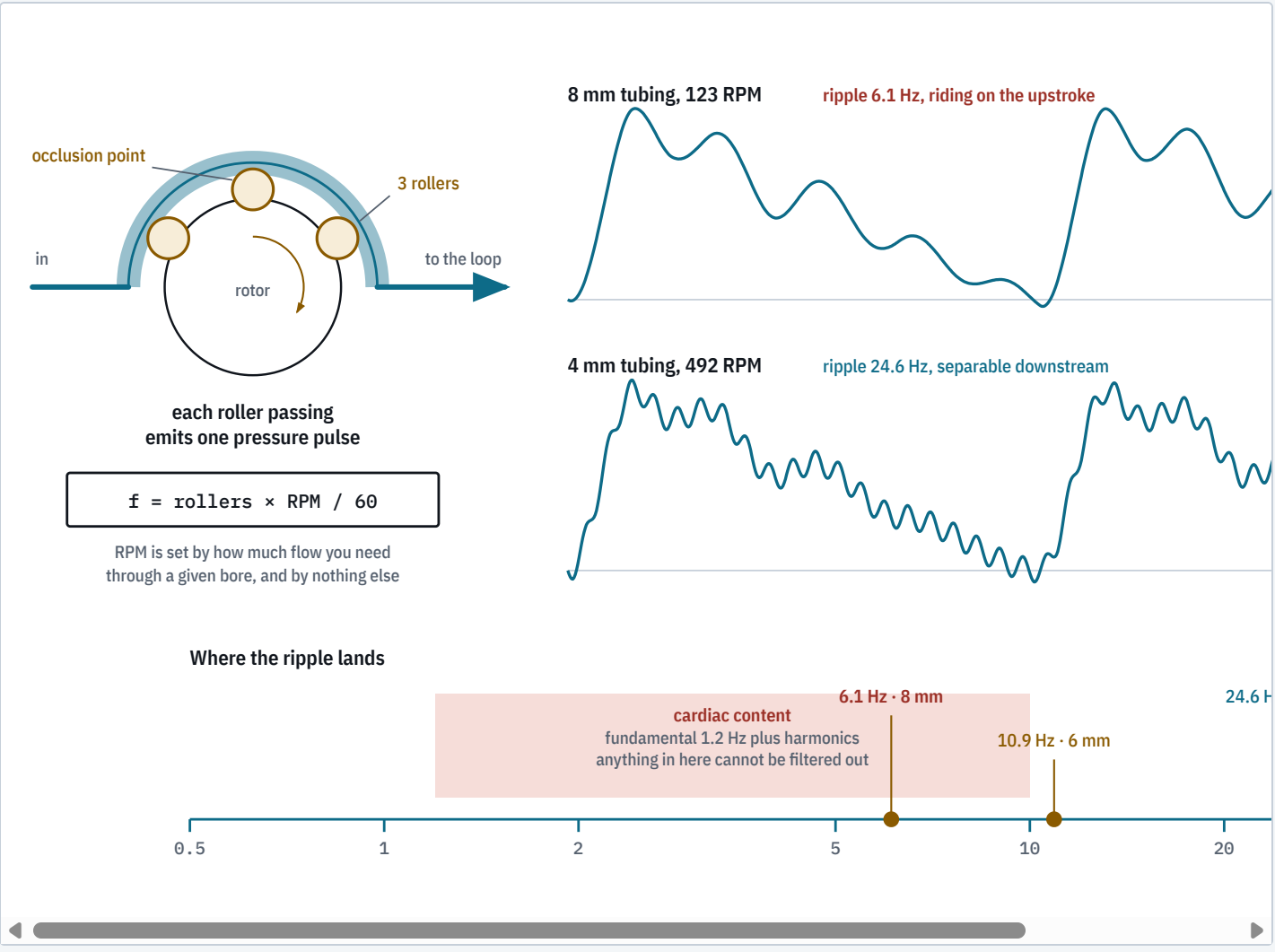
GW BME CAPSTONE · TCD PHANTOM · GROUP 2

# Pump and Target Waveform

Two things to settle before any hardware is ordered. What a peristaltic pump actually puts into the line, and what the loop has to produce at the middle cerebral artery. Nothing here needs the drive electronics built.

## 1 What the pump emits

A peristaltic produces flow plus a ripple at its own frequency, which has nothing to do with heart rate.




**The ripple has to sit clear of the cardiac band, and bore is the only knob that moves it.** Larger tubing means fewer revolutions for the same flow, which drags the ripple down into the frequency content of the systolic upstroke, where no downstream filtering can separate it again.

Drawing scrolls sideways.

| HEAD TUBING | RPM AT 467 ML/MIN | RIPPLE, 3 ROLLERS | VERDICT |
|---|---|---|---|
| 8.0 mm | 123 RPM | 6.1 Hz | inside the cardiac band |
| 6.0 mm | 218 RPM | 10.9 Hz | marginal, just clear |
| 4.0 mm | 492 RPM | 24.6 Hz | clear, filterable |

Scaled from one published OMEGAFLEX point, 2280 mL/min at 600 RPM on 8 mm, assuming flow goes as the square of bore. Check against the real flow curve, and count the rollers on the head, because that number sets the ripple directly and is not in your spreadsheet.

**A peristaltic is a good flow source and a bad waveform generator.**

Every design in your reading list separates those two jobs. Greaby used a metering pump and shaped the waveform with an air plenum downstream. Rominger used a constant flow pump with a separate downstream modulator. Making the pulse by speeding the pump up and slowing it down means fighting rotor inertia and the squeezed tubing, and the sharp systolic upstroke is the first thing you lose.

**Tapering after the pump does not help the ripple.**

Flow is conserved, so the pump still moves the same 467 mL/min and its RPM is set by the tubing in the head. Running 8 mm through the head and necking down at the phantom still leaves you at 123 RPM and 6.1 Hz. Two separate decisions.

**A more expensive pump does not fix this, and there is direct evidence.**

Roldan and Kyriacou built a head phantom in 2023 using a Harvard Apparatus 55-3305, a purpose-built pulsatile blood pump with adjustable stroke volume, rate and systolic fraction. Their own stated limitation is that *the resultant waveform does not have a clear dicrotic notch.* The notch is not something a pump strokes out. It comes from wave reflection against the impedance downstream of it, which means it is made by the plenum and the resistance or it is not made at all. Spend the effort there.

**Count the rollers before trusting any of this.**

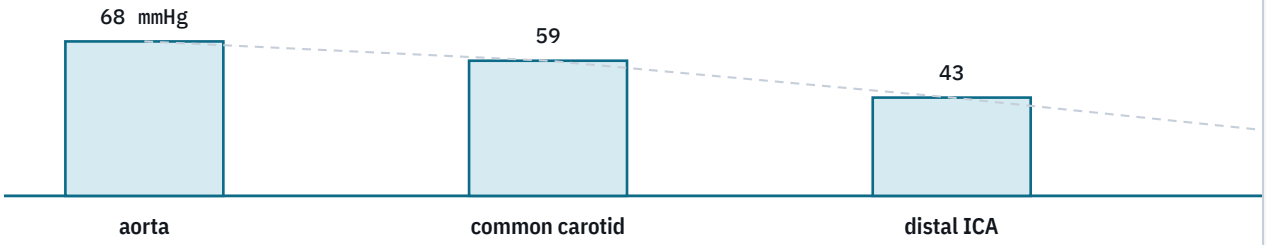
Three rollers at 492 RPM gives 24.6 Hz. Two rollers gives 16.4 Hz, six gives 49.2 Hz. It is a linear multiplier on the whole argument and it takes ten seconds to check.

# 2 What it has to produce

The cerebral bed never sees systemic pressure, which makes this easier to build than it first looks.

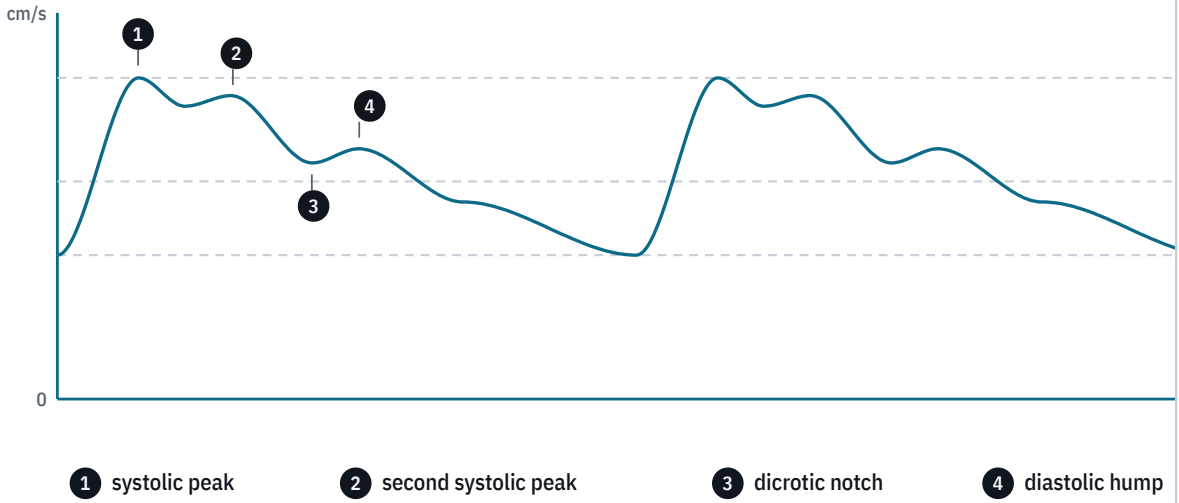
**Pulse pressure along the path, measured**

Shoemaker 2025, pressure wire, n = 5. A direction, not a specification.




63% of the pulse pressure is gone before it reaches you

**The velocity waveform at M1**




1 systolic peak  2 second systolic peak  3 dicrotic notch  4 diastolic hump

**High diastolic velocity is the feature that makes this a cerebral waveform.** The brain is a low-resistance bed, so flow never stops between beats. A peripheral artery decays toward zero in diastole and a carotid sits in between. Reproducing that sustained diastolic floor is the real test of whether compliance and resistance are set correctly.

**Your own number was right, and it changes the build.**

You wrote that cerebral pulse pressure is heavily damped inside the skull, 20 to 30 mmHg against 60 plus systemically. That matches the measured values above. The loop does not have to reproduce systemic pressures, which lowers what both the pump and the vessel have to survive.

**Pulsatility index ties the three velocity targets together.**

PI is (PSV minus EDV) divided by MFV. Using the midpoints of your own bands, 100, 45 and 67.5, that gives

**PI about 0.8**

, which sits right on published normals for a healthy MCA. If the rig produces the right peak but PI comes out high, diastole is collapsing, and the cause is almost always too much resistance or too little compliance.

**Call the landmarks by their velocity names.**

P1, P2 and P3 are intracranial pressure nomenclature and mean something else entirely. On a velocity trace they are the systolic peak, second systolic peak, dicrotic notch and diastolic hump.

**The pump sets the mean, the plenum and the clamp set the shape.**

Compliance stores volume and gives it back, which is what fills in diastole. Resistance restricts flow and sets how fast pressure decays. They are independent knobs and the whole design depends on turning them separately. A plenum is compliance. A downstream constriction is resistance. Wall stiffness is compliance, not resistance, even though the word feels right for a stiff material.

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Velocity and diameter bands are Group 2's own, from the pump comparison sheet. Pressure values are Shoemaker 2025. Ripple figures are derived rather than measured and want checking on the bench. If a number here disagrees with `capstone/INTERFACE-CONTROL-DOCUMENT.md`, that file wins, and tell me.